

QA Portfolio

DevelopersHubCorporation Internship | Domain: Software Quality Assurance (SQA)

Intern: Esha Tabbassum | ID: DHC-1019

Duration: 6 Weeks (February – March 2026)

Project: Bug Tracking and Test Automation for Web Application

Application Under Test: SauceDemo (<https://www.saucedemo.com/>)

1. About Me

I am a Software Quality Assurance intern with a strong passion for ensuring software quality through structured testing. Over the past 5 weeks, I have gained hands-on experience in manual testing, automation, performance testing, and security testing. This portfolio documents my complete journey.

2. Skills Demonstrated

Category	Skills / Tools
Manual Testing	Test case design, execution, bug reporting
Bug Tracking	Jira
Automation	Selenium WebDriver, Python
Performance Testing	Apache JMeter 5.6
Security Testing	OWASP ZAP 2.17.0
Version Control	GitHub
CI/CD (Theory)	GitHub Actions
Documentation	Microsoft Word

3. Week-by-Week Summary

Week 1 – Manual Testing & Bug Reporting

- Created **8 test cases** for login functionality
- Identified **3 functional/UI bugs**
- Reported bugs using **Jira**
- **Deliverables:** Test Case Document, Bug Report Sheet

Week 2 – Test Planning & Advanced Testing

- Developed comprehensive **Test Plan**
- Applied **BVA** and **EP** black-box techniques
- Performed **cross-browser testing** (Chrome, Firefox, Edge)
- Retested Week 1 bugs
- **Deliverables:** Test Plan, Compatibility Report

Week 3 – Automation Testing

- Automated **login validation** using Selenium
- Automated **product sorting (Z-A)** using Selenium
- Captured **screenshots** as evidence
- **Deliverables:** Python scripts (`login\_test.py`, `dashboard\_test.py`), screenshots, Final Report

Week 4 – Performance & Load Testing

- Created load test plan with **50–100 virtual users**
- Executed tests using **Apache JMeter 5.6**
- Analyzed response time, throughput, and error rate
- **Deliverables:** Performance Test Plan, Performance Test Report

Week 5 – Security Testing

- Performed security scan using **OWASP ZAP 2.17.0**
- Identified **12 security findings** (4 Medium, 4 Low, 4 Informational)
- Documented each vulnerability with evidence and solution
- **Deliverables:** Security Testing Summary, Vulnerability Report

4. Key Achievements

Area	Achievement
Manual Testing	Found and documented 3 critical UI/UX bugs
Automation	Created 2 fully functional Selenium scripts with 100% pass rate
Performance	Successfully tested SauceDemo with up to 100 concurrent users
Security	Identified 12 security vulnerabilities including missing CSP, CORS misconfiguration, and information disclosure
Documentation	Produced 10+ professional QA reports (PDF format)

5. Sample Work

Bug Report Example

Bug ID: BUG-001

Title: Mobile error message overlaps password field

Severity: Medium

Steps to Reproduce:

1. Open Chrome DevTools → iPhone X view
2. Leave username blank
3. Click Login
4. Observe error message overlapping password field

Automation Script Example

login_test.py

```
from selenium import webdriver

from selenium.webdriver.common.by import By
import time

driver = webdriver.Chrome()

driver.get("https://www.saucedemo.com/")

driver.find_element(By.ID, "user-name").send_keys("standard_user")
```

```
driver.find_element(By.ID, "password").send_keys("secret_sauce")
driver.find_element(By.ID, "login-button").click()
time.sleep(3)
```

```
if "inventory" in driver.current_url:
    print("✅ Login successful")
    driver.save_screenshot("login_success.png")
driver.quit()
```

Security Finding Example

Finding: Content Security Policy (CSP) Header Not Set

Risk: Medium

Description: CSP helps prevent XSS attacks by specifying allowed content sources.

Solution: Configure server to include Content-Security-Policy header.

6. Testing Summary

Week	Focus	Key Deliverables	Status
Week 1	Manual Testing	Test Cases (8) + Bug Report (3 bugs)	Done
Week 2	Test Planning	Test Plan + Compatibility Report	Done
Week 3	Automation	2 Python scripts + Final Report	Done
Week 4	Performance	Test Plan + Test Report (50–100 users)	Done
Week 5	Security	Security Summary + Vulnerability Report (12 findings)	Done

7. Lessons Learned

- How to design effective and structured test cases
- How to track and document bugs using Jira
- How to automate web applications using Selenium WebDriver

- How to perform load testing with Apache JMeter
- How to identify security vulnerabilities using OWASP ZAP
- How to document and present QA findings professionally

8. Recommendations for Future Work

1. Fix identified bugs (mobile overlap, forgot password, password toggle)
2. Expand automation coverage to include checkout process
3. Implement CI/CD pipeline for automated test execution
4. Perform load testing with 200+ users to find breaking points
5. Integrate security testing into development lifecycle

9. Conclusion

This 5-week internship successfully covered all major phases of software quality assurance:

- Manual testing foundation
- Test planning and advanced techniques
- Automation testing with Selenium
- Performance testing with JMeter
- Security testing with OWASP ZAP

All deliverables were completed on time, documented professionally, and uploaded to GitHub for reference.

10. Contact & Portfolio

GitHub Repository: <https://github.com/esha-tabassum/SQA-Internship-Project>

Email: [eshatabassum46@gmail.com]

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This portfolio documents my 6-week journey from manual testing to security testing.

